

EENG250 ELECTRIC CIRCUITS I · CHAPTER 5

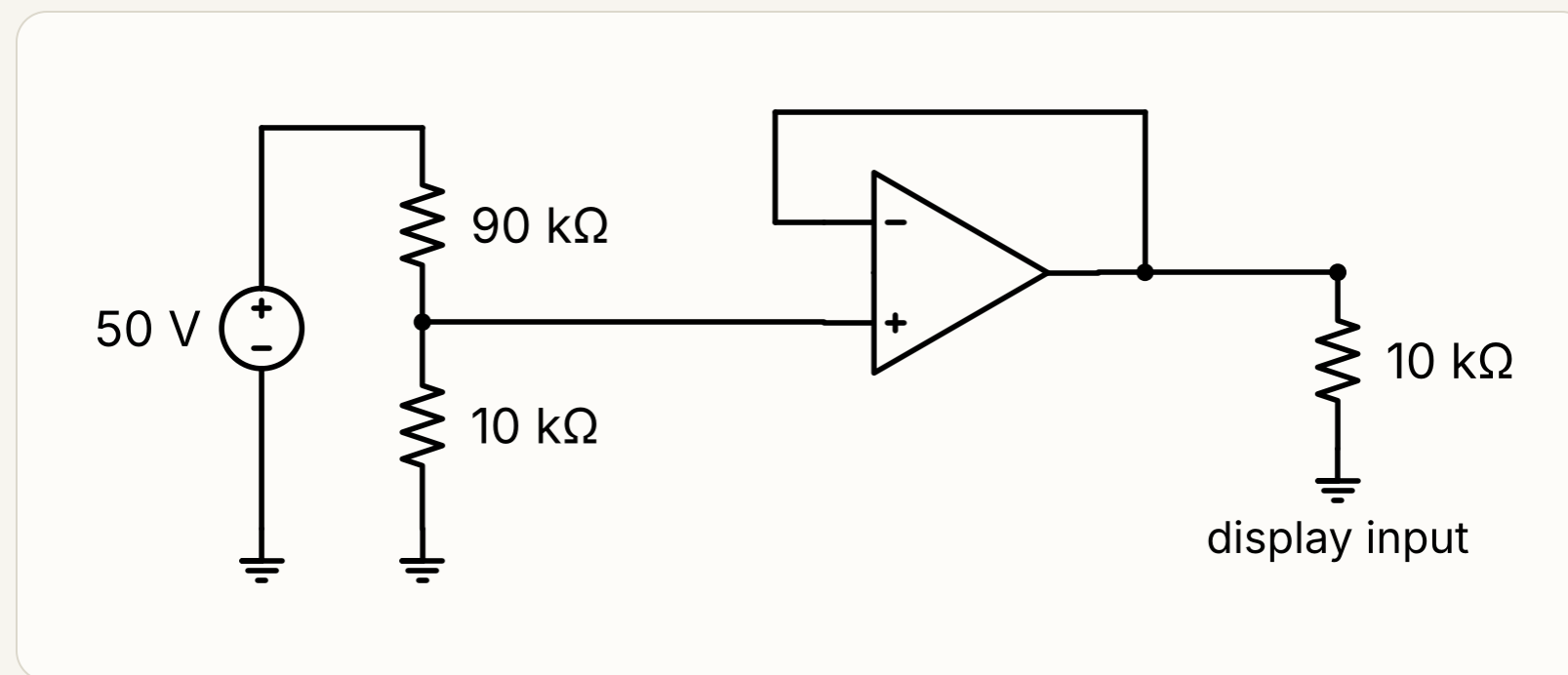
The Operational Amplifier

Two rules and one node equation give circuits that amplify, add, subtract and buffer signals.

Prepared by Dr. Adel Daaboul

PREDICT FIRST

A divider scales 50 V down to 5 V for a display. Without the op amp, what does the display read?



5 V

about 4.5 V

about 2.6 V

Without the op amp, the display's 10 kΩ input sits straight across the divider's 10 kΩ.

CHAPTER 5 GOALS

By the end of Chapter 5 you can

1 • The ideal op amp

State the two rules, when they hold, and what happens when the output saturates.

2 • Four amplifiers

Name the inverting, summing, noninverting and difference amplifiers and write v_o for each.

3 • The linear range

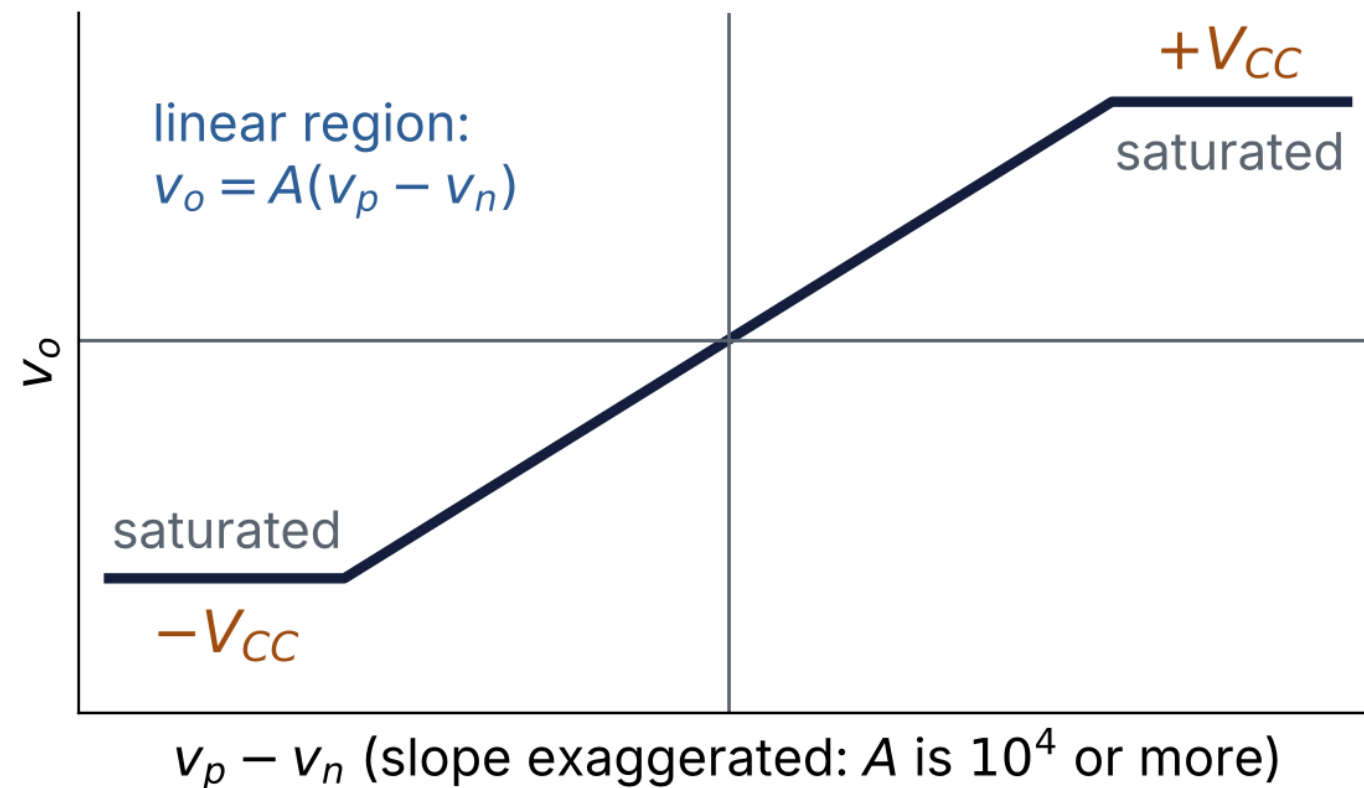
Find the inputs that keep the op amp linear, and the output when it is not.

4 • Loads and cascades

Find the output current, and analyse two stages in a row.

TERMINALS AND TRANSFER CHARACTERISTIC

The output is trapped between the supplies



Five terminals that matter: the inverting input (v_n), the noninverting input (v_p), the output (v_o) and the supplies $+V_{CC}$ and $-V_{CC}$.

Linear region: $v_o = A(v_p - v_n)$. The gain A is 10 000 or more, so about a millivolt or less between the inputs drives the output to a supply.

Saturated: v_o sits at $+V_{CC}$ or $-V_{CC}$.

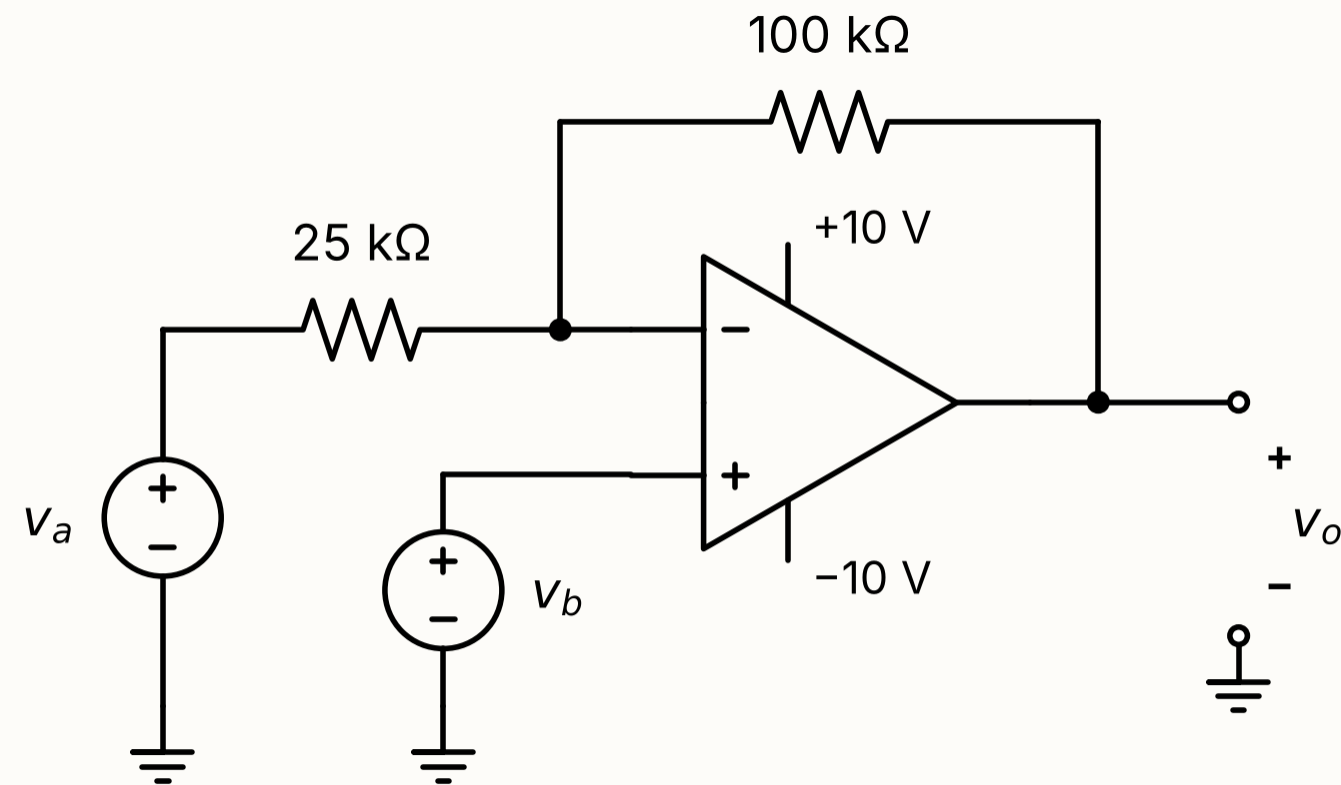
With negative feedback, two rules replace the op amp

$$i_p = i_n = 0, \quad v_p = v_n \quad (\text{negative feedback, linear region})$$

- 1 Check for negative feedback: a path from the output back to the inverting input.
- 2 Set $v_n = v_p$ and $i_n = i_p = 0$.
- 3 Write KCL at the inverting input: the sum of the currents leaving it is zero.
- 4 Solve for v_o , then check that it lies between the supplies. If not, the op amp saturates.

NILSSON EXAMPLE 5.1

Both inputs driven: $v_n = v_b$



Supplies $\pm 10\text{ V}$. Find v_o for:

a) $v_a = 1\text{ V}$ and $v_b = 0\text{ V}$

b) $v_a = 1\text{ V}$ and $v_b = 2\text{ V}$

c) $v_a = 1.5\text{ V}$: the range of v_b that avoids saturation

a) KCL at the inverting input, currents leaving:

$$v_n = v_b = 0 : \quad \frac{0 - 1}{25} + \frac{0 - v_o}{100} = 0 \Rightarrow v_o = -4\text{ V}$$

NILSSON EXAMPLE 5.1

v_n follows v_p , so the inverting input is no longer at 0 V

b) Same KCL, with $v_n = v_b = 2$ V:

$$v_n = v_b = 2 \text{ V} : \quad \frac{2 - 1}{25} + \frac{2 - v_o}{100} = 0 \quad \Rightarrow \quad v_o = 6 \text{ V}$$

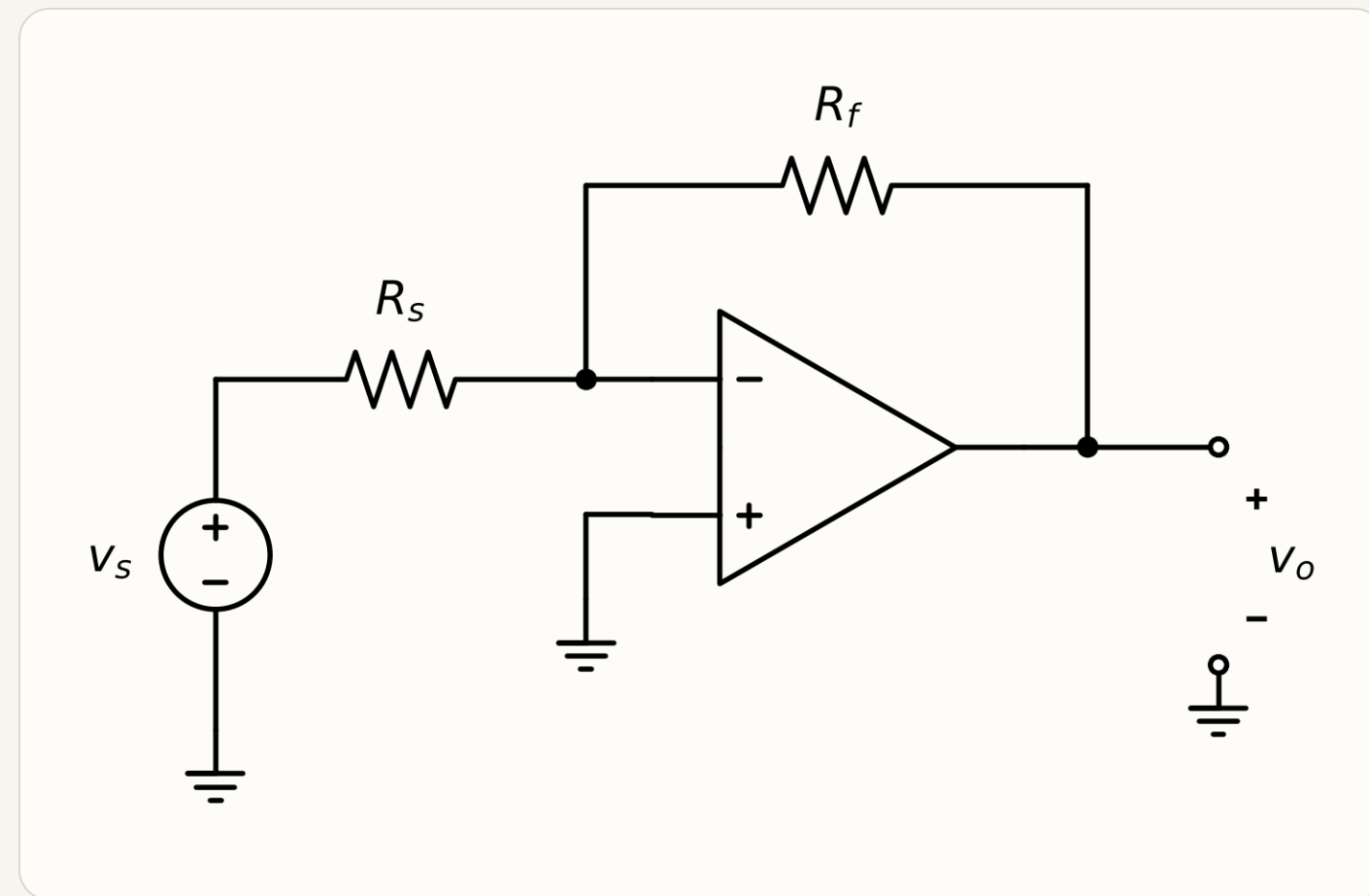
Check (b): with $v_o = 6$ V, $(2 - 1)/25 + (2 - 6)/100 = 0.04 - 0.04 = 0$. KCL holds.

c) In general, the KCL gives $v_o = 5v_b - 4v_a$. Keep v_o between the supplies:

$$v_o = 5v_b - 4v_a = 5v_b - 6, \quad -10 \leq v_o \leq 10 \quad \Rightarrow \quad -0.8 \text{ V} \leq v_b \leq 3.2 \text{ V}$$

THE INVERTING AMPLIFIER

The gain is set by two resistors



$$v_o = -\frac{R_f}{R_s} v_s$$

$v_p = 0$, so $v_n = 0$: a virtual ground. The current v_s/R_s flows on through R_f .

Nilsson Example 5.2: a gain of 12 with ± 15 V supplies. Choose $R_s = 1$ k Ω and $R_f = 12$ k Ω ; then find the linear range of v_s :

$$v_o = -\frac{12}{1} v_s = -12v_s, \quad |v_o| \leq 15 \Rightarrow -1.25 \text{ V} \leq v_s \leq 1.25 \text{ V}$$

PREDICT FIRST

The amplifier of Example 5.2 gets $v_s = 2 \text{ V}$. What is v_o ?

−24 V

−15 V

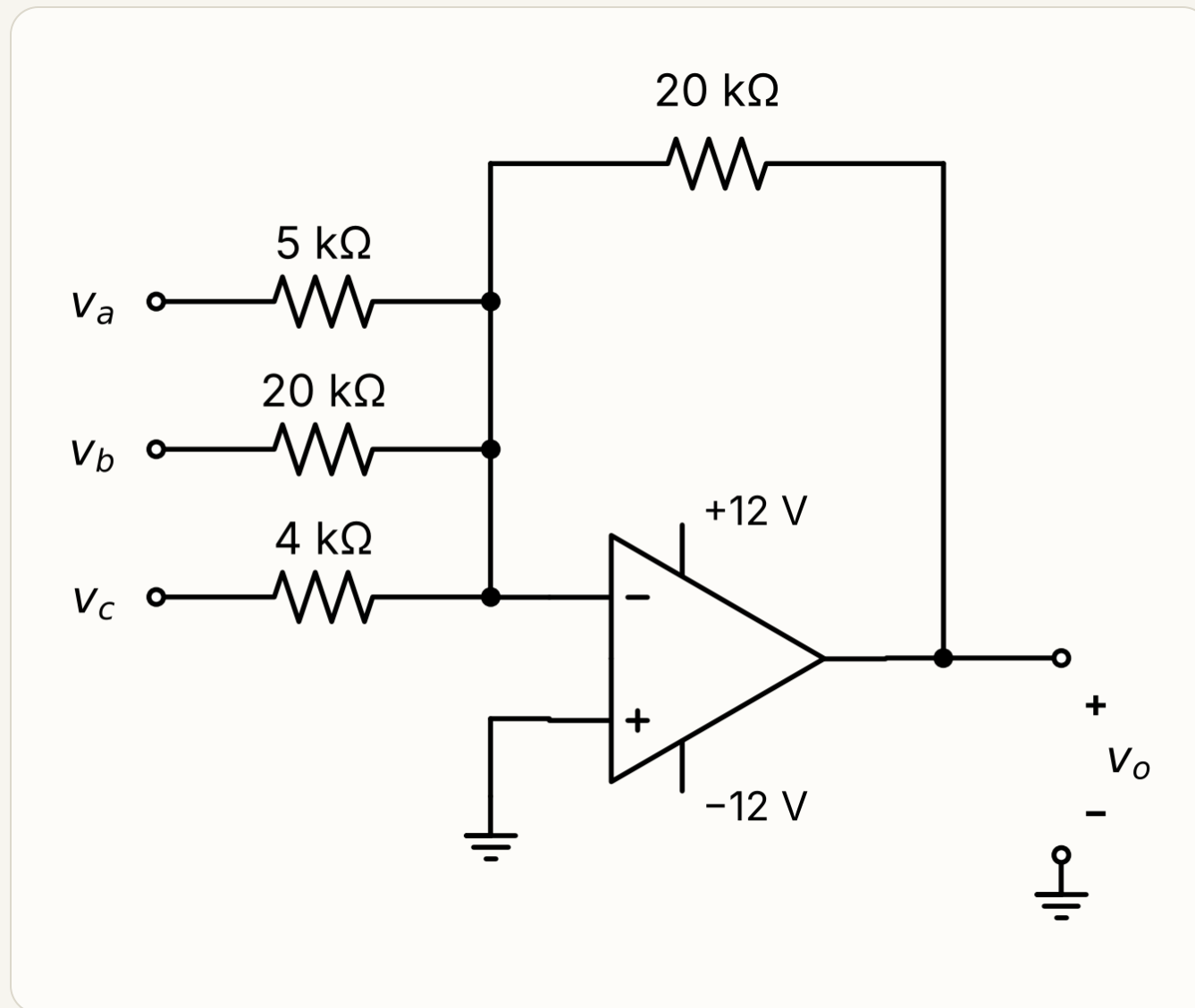
+15 V

0 V

$$v_s = 2 \text{ V} : \quad -12(2) = -24 \text{ V is beyond } -15 \text{ V} \Rightarrow v_o = -15 \text{ V}$$

−15 V. The output cannot pass a supply. Once it sits there, the feedback can no longer hold $v_n = v_p$: here $v_n = 0.69 \text{ V}$, not 0. Only $i_n = i_p = 0$ still holds.

Each input adds its own term



$$v_o = - \left(\frac{R_f}{R_a} v_a + \frac{R_f}{R_b} v_b + \frac{R_f}{R_c} v_c \right)$$

Design $v_o = -(4v_a + v_b + 5v_c)$ with $R_f = 20 \text{ k}\Omega$ and $\pm 12 \text{ V}$ supplies.

a) Choose the input resistors:

$$\frac{20}{R_a} = 4, \quad \frac{20}{R_b} = 1, \quad \frac{20}{R_c} = 5 \Rightarrow R_a = 5, \quad R_b = 20, \quad R_c = 4 \text{ k}\Omega$$

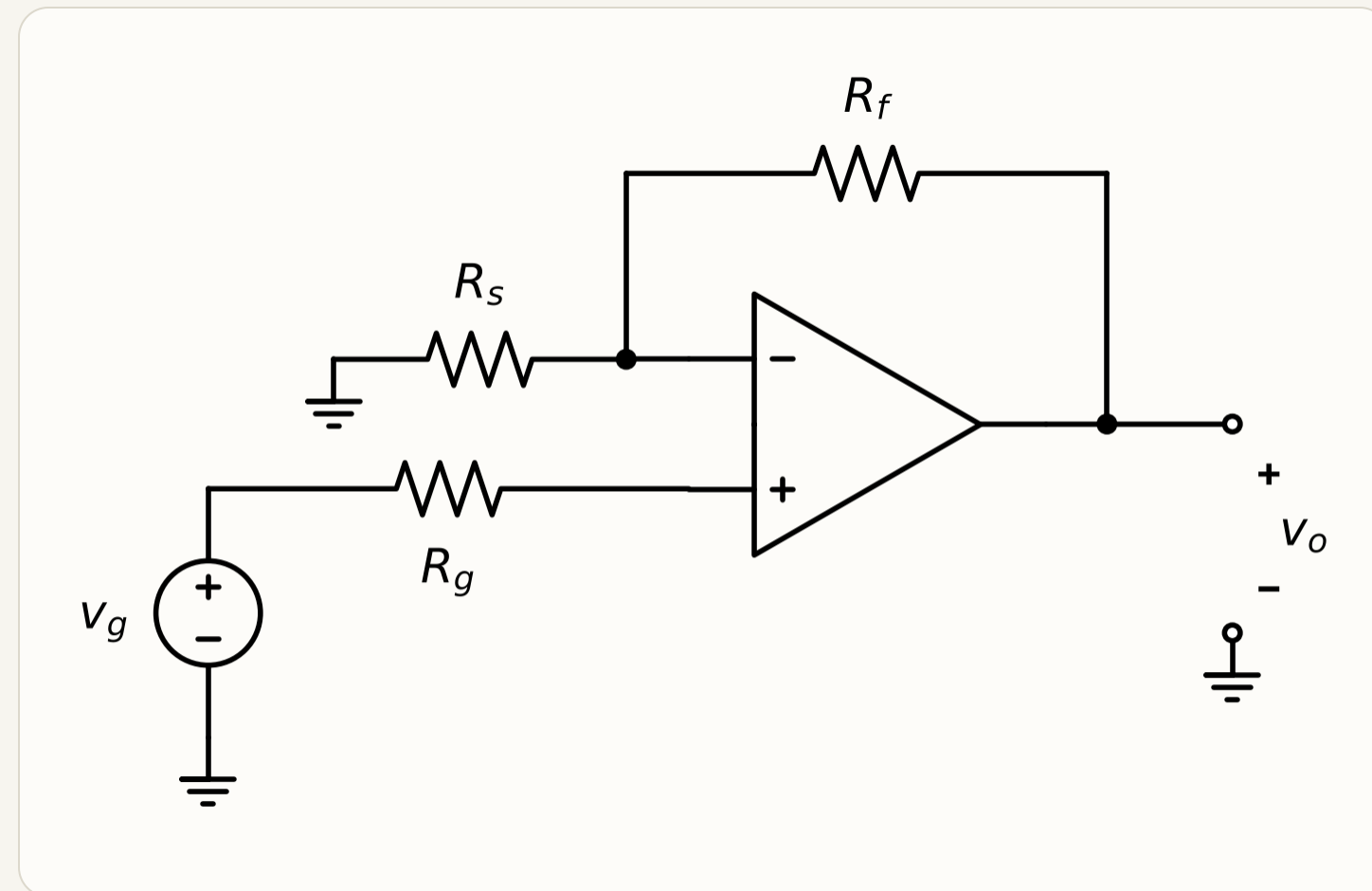
b) With $v_a = 2 \text{ V}$ and $v_c = -1 \text{ V}$, what range of v_b keeps the op amp linear?

$$v_a = 2, \quad v_c = -1 : \quad v_o = -8 - v_b + 5 = -3 - v_b \Rightarrow -15 \text{ V} \leq v_b \leq 9 \text{ V}$$

c) With $v_a = 2 \text{ V}$, $v_b = 3 \text{ V}$ and $v_c = -1 \text{ V}$, how large can R_f be?

$$v_o = -R_f \left(\frac{2}{5} + \frac{3}{20} - \frac{1}{4} \right) = -0.3R_f = -12 \Rightarrow R_f = 40 \text{ k}\Omega$$

The source drives the noninverting input



$$v_o = \left(1 + \frac{R_f}{R_s} \right) v_g$$

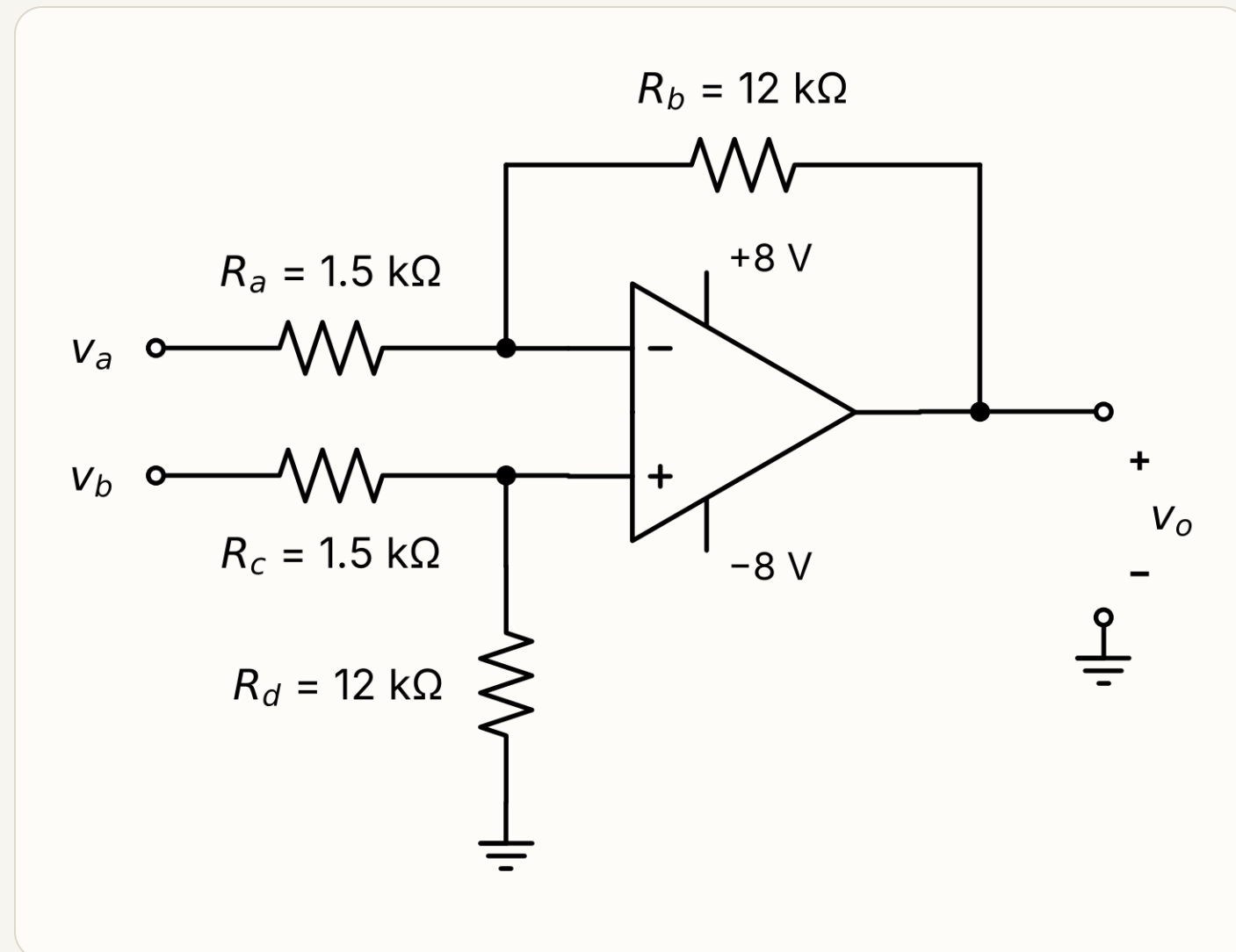
No current enters the input, so R_g drops no voltage: $v_p = v_g$. The divider gives $v_n = v_o R_s / (R_s + R_f)$, and $v_n = v_p = v_g$.

$R_f = 0$: the voltage follower, $v_o = v_g$.

Example 5.4: a gain of 6 ($R_s = 2 \text{ k}\Omega$, $R_f = 10 \text{ k}\Omega$, $R_g = 1 \text{ k}\Omega$); v_g from -1.5 V to 1.5 V . The smallest supplies:

$$1 + \frac{10}{2} = 6, \quad v_o = 6(\pm 1.5) = \pm 9 \text{ V} \Rightarrow \pm 9 \text{ V supplies}$$

It amplifies $v_b - v_a$



$$v_o = \frac{R_b}{R_a} (v_b - v_a) \quad \text{when} \quad \frac{R_a}{R_b} = \frac{R_c}{R_d}$$

Example 5.5: a gain of 8 with $\pm 8 \text{ V}$ supplies. Choose $R_a = R_c = 1.5 \text{ k}\Omega$ and $R_b = R_d = 12 \text{ k}\Omega$.

With $v_a = 1 \text{ V}$, what range of v_b keeps the op amp linear?

$$v_o = \frac{12}{1.5} (v_b - v_a) = 8(v_b - 1), \quad |v_o| \leq 8 \Rightarrow 0 \leq v_b \leq 2 \text{ V}$$

WHEN THE RATIOS DO NOT MATCH

Find v_p from the divider, then KCL at v_n

$$v_p = \frac{R_d}{R_c + R_d} v_b, \quad v_o = \left(1 + \frac{R_b}{R_a}\right) v_p - \frac{R_b}{R_a} v_a$$

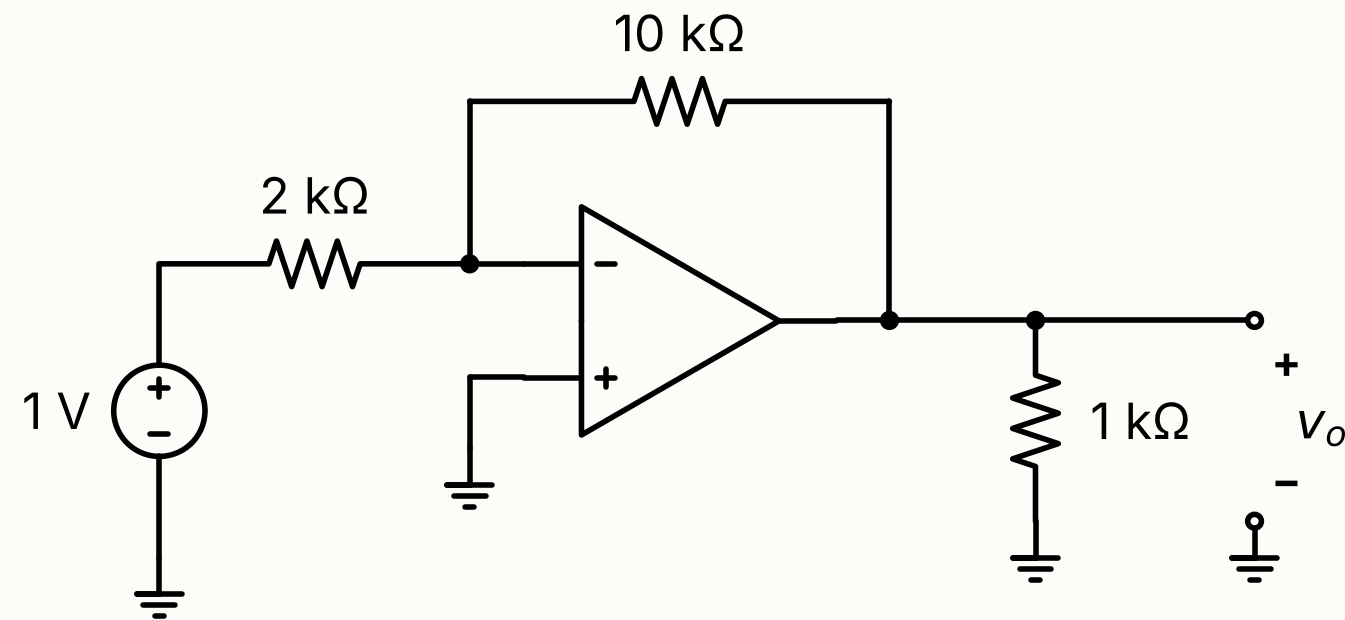
This holds for any resistors. The short formula $(R_b/R_a)(v_b - v_a)$ needs $R_a/R_b = R_c/R_d$.

Example

$R_a = R_c = 10 \text{ k}\Omega$, $R_b = 40 \text{ k}\Omega$, $R_d = 10 \text{ k}\Omega$, $v_a = 1 \text{ V}$, $v_b = 2 \text{ V}$: $v_p = 1 \text{ V}$, so $v_o = 5(1) - 4(1) = 1 \text{ V}$, not $4(2 - 1) = 4 \text{ V}$.

THE OUTPUT CURRENT

The inputs draw no current. The output can.



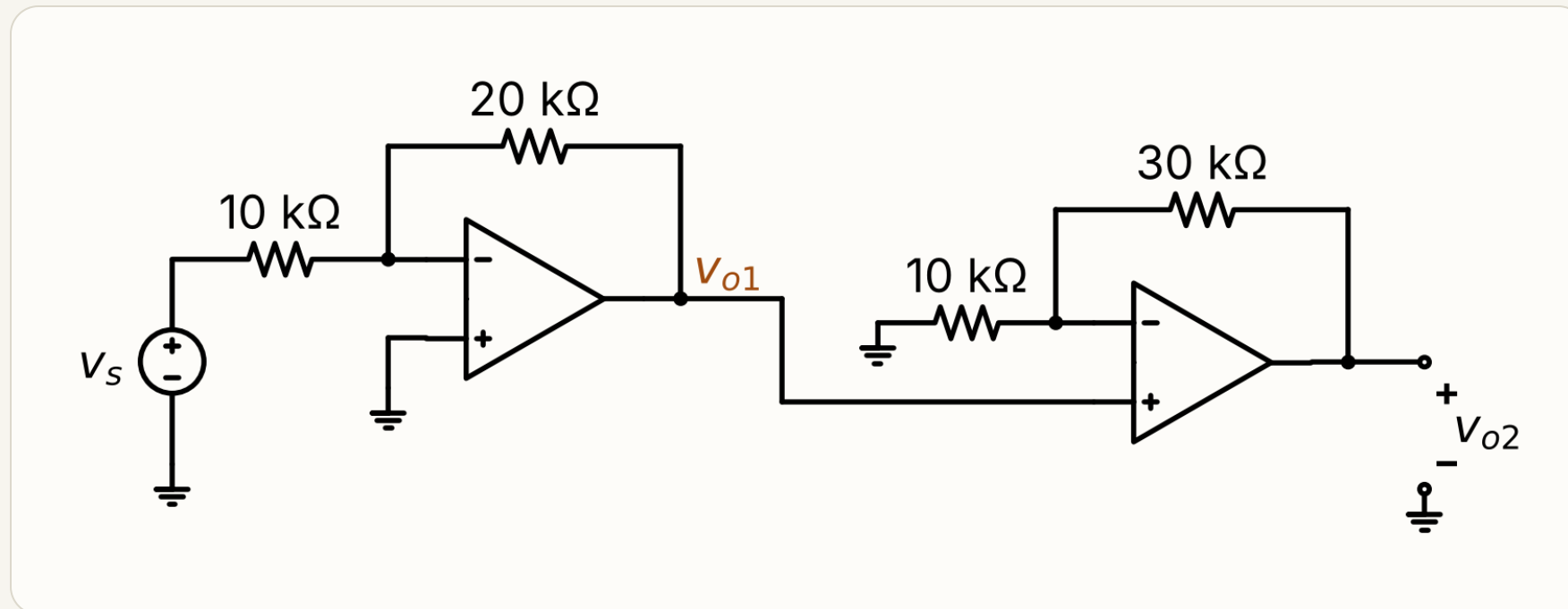
With $v_o = -5$ V, current flows up through the 1 kΩ load and through the 10 kΩ, both into the output node. The output stage sinks the 5.5 mA into the negative supply.

KCL at the output node, not at an input:

$$v_o = -5 \text{ V}, \quad i_L = \frac{5}{1} = 5 \text{ mA up}, \quad i_f = \frac{0 - (-5)}{10} = 0.5 \text{ mA} \Rightarrow 5.5 \text{ mA into the output}$$

TWO OP AMPS IN CASCADE

Gains multiply; every stage must stay linear



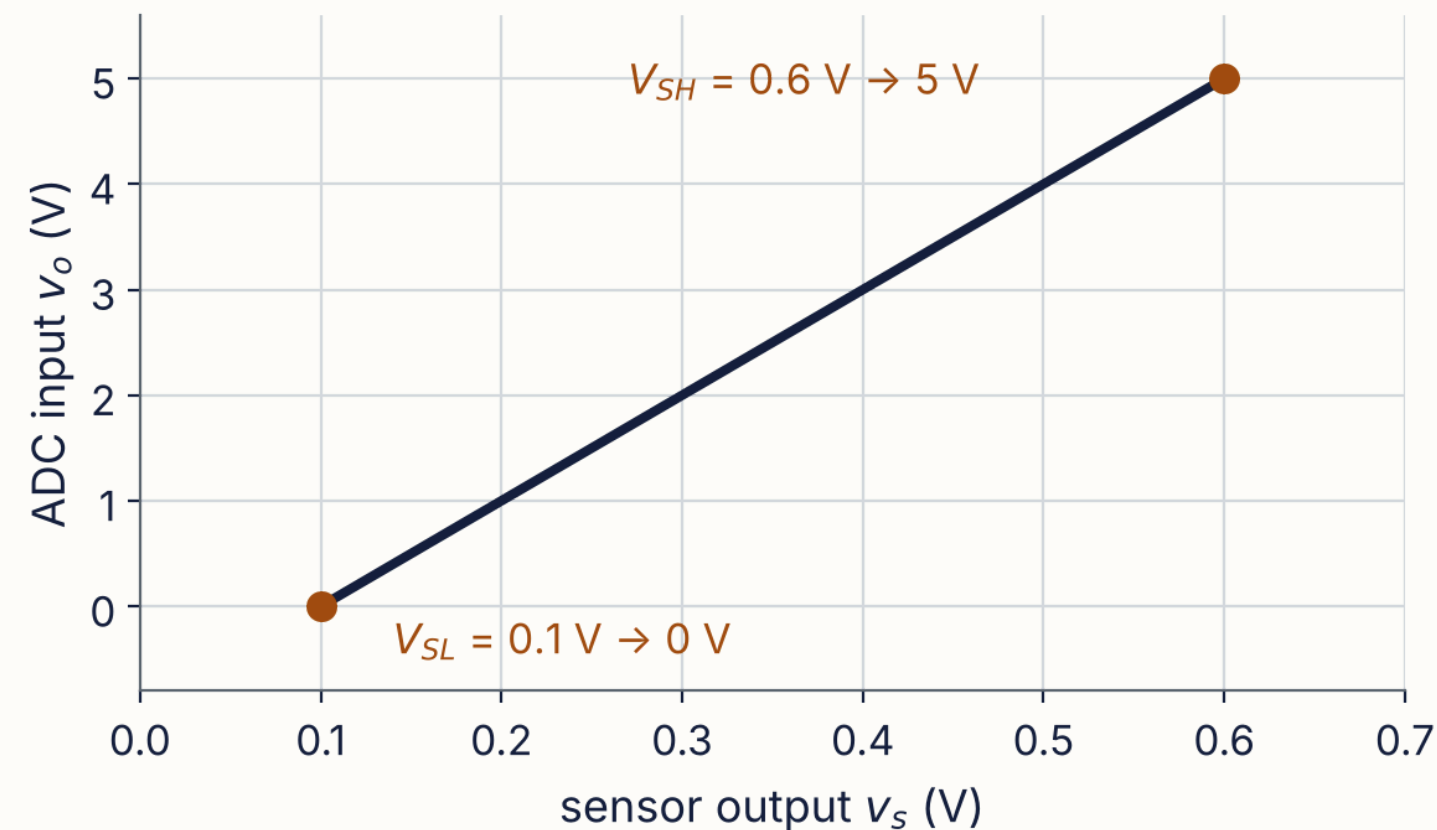
Both op amps on ± 12 V. The second stage's input draws no current, so it does not load the first:

$$v_{o1} = -2v_s, \quad v_{o2} = \left(1 + \frac{30}{10}\right) v_{o1} = -8v_s, \quad |v_{o2}| \leq 12 \Rightarrow |v_s| \leq 1.5 \text{ V}$$

Stage 1 alone allows $|v_s| \leq 6$ V; stage 2 saturates first. Check every stage, not only the last.

PRACTICAL PERSPECTIVE: SENSORS

Map a sensor's range onto an ADC's input range



A sensor gives 0.1 V to 0.6 V; the analog-to-digital converter reads 0 V to 5 V. Use the full range with $v_o = Kv_s + L$.

K amplifies; L shifts the level. An op amp with a reference voltage does both (Nilsson Fig. 5.23).

$$K = \frac{V_{OH} - V_{OL}}{V_{SH} - V_{SL}} = \frac{5 - 0}{0.6 - 0.1} = 10, \quad L = V_{OL} - K V_{SL} = -1 \text{ V}$$

CHECKING AN AI ANSWER

An AI solved Example 5.1(b) like this. What is wrong?

$$i = \frac{2 - 1}{25} = 0.04 \text{ mA from } v_a \text{ into the node, } v_o = 2 - (0.04)(100) = -2 \text{ V}$$

Correct:

0.04 mA leaves the node toward v_a , so it arrives from the output: $v_o = 2 + (0.04)(100) = 6 \text{ V}$

With $v_n = 2 \text{ V}$ and $v_a = 1 \text{ V}$, the current flows out of the node toward v_a . The giveaway: KCL at the inverting input fails with the AI's v_o .

$$\frac{2 - 1}{25} + \frac{2 - (-2)}{100} = 0.08 \text{ mA} \neq 0$$

In a 2025 study of Gemini 2.5 Pro on undergraduate circuit problems, misread source polarities caused about a third of its wrong answers, and misread current directions nearly as many.

THE FOUR CONFIGURATIONS AND THE FOLLOWER

Name it first, then write v_o

Configuration	v_o	Recognise it by
Inverting	$-(R_f/R_s)v_s$	Source to the inverting input; noninverting input grounded
Summing	$-(R_f/R_a)v_a - (R_f/R_b)v_b - \dots$	Several inputs to the inverting input
Noninverting	$(1 + R_f/R_s)v_g$	Source to the noninverting input; R_s from the inverting input to ground
Follower ($R_f = 0$)	v_g	Output wired straight to the inverting input
Difference	$(R_b/R_a)(v_b - v_a)$	One input on each side; only when $R_a/R_b = R_c/R_d$

CHECK EVERY ANSWER

Three checks, no second solution needed

$$\sum i \text{ leaving } v_n = 0, \quad -V_{CC} \leq v_o \leq V_{CC}$$

KCL at v_n

Put your v_o back: the currents leaving the inverting input must sum to zero.

Between the supplies

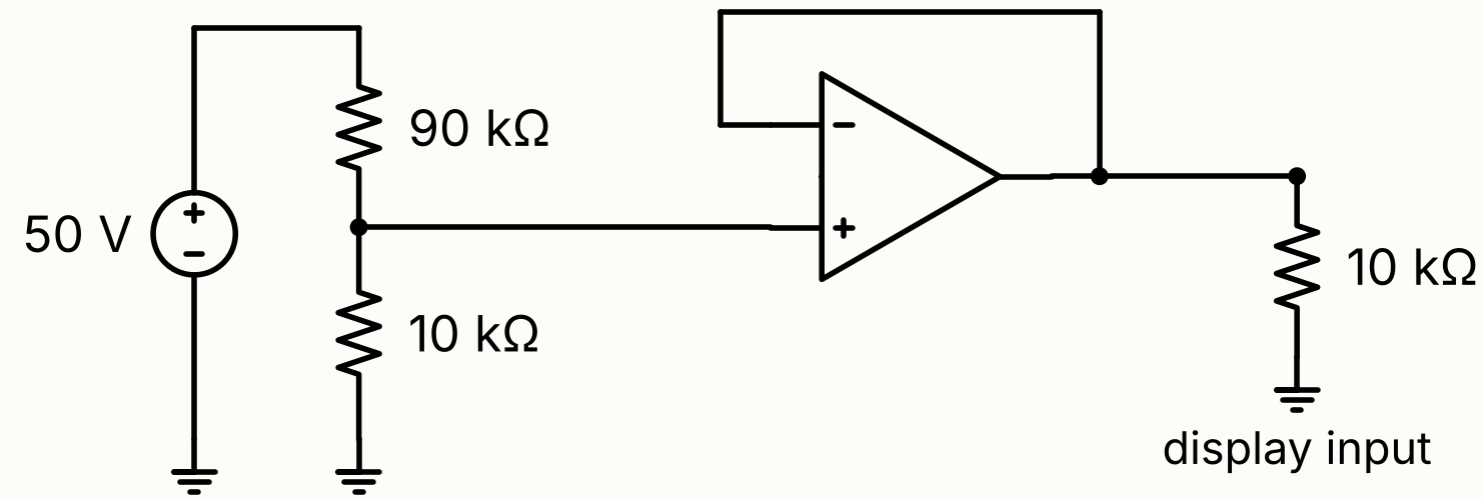
An output beyond $\pm V_{CC}$ is not an answer: the op amp saturates there.

The sign

An inverting stage flips the sign of its input; a noninverting stage keeps it.

BACK TO THE BATTERY MONITOR

A follower stops the display loading the divider



$$\text{no op amp: } 10 \parallel 10 = 5 \text{ k}\Omega, \quad v = 50 \frac{5}{90 + 5} = 2.63 \text{ V}$$

$$\text{follower: } v_p = 50 \frac{10}{90 + 10} = 5 \text{ V}, \quad v_o = v_p = 5 \text{ V}$$

The follower's input draws no current, so the divider is unloaded. Its output delivers the display's 0.5 mA, drawn from the op amp's supply rails.

WHAT COMES NEXT

Practise openly, then show it on your own

In class

Booklet Problems 5.1 to 5.11.
Problem 5.12 is homework.

At home

The 16-item practice sheet. AI
tools allowed; not marked.

Week 11

15-minute quiz, no devices.
Every question is a twin of a
practice-sheet question.

Chapter 6: inductors and capacitors. A capacitor in the feedback path makes the op amp integrate its input.

CHECKLIST

You can do these without help

- State the two ideal op-amp rules and when they apply
- Name an amplifier's configuration from its circuit and write v_o in terms of its inputs
- Analyse the inverting, summing, noninverting and difference amplifiers with KCL at the inverting input
- Find the range of an input that keeps the op amp linear, and the output when it saturates
- Find the current in a load and at the op amp's output
- Analyse two op amps in cascade and say which stage saturates first